

Odor Classification using Machine Learning

Dhruti Avadhani

Aneesh Adithya SR

Buvica M

Gauravi Suryavamshi

Abstract—The classification of odors using machine learning and sensor-based technologies presents a frontier in the intersection of artificial intelligence and chemical sensing. In this project, we propose an odor classification framework leveraging synthetic datasets generated from known olfactory-active compounds and their corresponding detection thresholds. With the gas sensors MQ and TGS series in mind, we created a 16-feature dataset encompassing five distinct odor classes. Our approach involves training and evaluating multiple models such as Decision Trees, Random Forests, and Deep Neural Networks to identify the most effective methodology for odor recognition. We found that both Random Forests and DNNs outperformed traditional Decision Trees in terms of Precision, Recall and F1-score. This work contributes to the growing field of electronic noses (e-noses) and highlights the potential of integrating olfactory perception into AI systems, opening up new applications in healthcare diagnostics, environmental monitoring and multisensory AI development.

Index Terms—Odor Classification, Gas Sensors, MQ Sensors, TGS Sensors, Electronic Nose, Machine Learning, Deep Neural Network (DNN), Random Forest, Decision Tree, Volatile Organic Compounds (VOCs), Chemiresistive Sensing, Scent Detection, Synthetic Dataset, Multi-class Classification, TensorFlow, Keras.

I. INTRODUCTION

Odor Classification using sensors is an innovative approach that combines chemical sensing technologies with machine learning to identify and categorize various smells. This process typically relies on electronic noses (e-noses), which consist of an array of gas sensors such as the MQ and TGS sensors, which are primarily designed to detect Organic Compounds in the air. These sensors produce unique signal patterns in response to different odorants, enabling the extraction of features that can be analyzed computationally. In healthcare, odor-based diagnostics can help detect diseases such as diabetes or infections through breath analysis. Even in the fragrance and cosmetics industry, electronic noses are being developed to help design and replicate complex scent profiles.

Beyond practical applications, odor detection systems also hold potential in advancing the Turing Test - the evaluation of a machine's ability to exhibit intelligent behavior indistinguishable from that of a human. A machine can not only recognize visual and auditory cues but also "smell" and interpret scents contextually. Integrating olfaction into AI systems brings machines one step closer to human-like perception.

II. RELATED WORK

Some work has already been done in this field. Major companies like Aryballe (France) and Alpha MOS (France) focus on Electronic noses and chemical analysis

specifically for Food & Beverage Quality Control and Profiling. Osmo (acquired by Google DeepMind) also focuses on Olfactory perception modelling using machine learning, to map the relationship between molecular structure and human-perceived odor.

Apart from major companies, there has also been research in both the domains of odor detection using electronic noses and establishing a relationship between odor and molecular structure. Various synthetic datasets have been specifically developed for classifying odor based on sensor data. Other datasets such as SMILES and DREAM are developed exclusively for drawing a relationship between molecular structures and scents.

For the sensor based projects, Decision Trees and Random Forests have been implemented to train the model, whereas for the projects focused on molecular structure, Deep Neural Networks (DNN), Convolution Neural Networks (CNN), Graphical Neural Networks (GNN) and Graphical Convolution Neural Networks (GCN) have been used.

III. METHODOLOGY

We first curated our own synthetic dataset for this project. Certain chemicals and organic compounds have a distinctive smell of their own and this smell can be detected when it is present in the atmosphere in certain quantities (ppm - parts per million). Research data clearly establishes these ranges where these chemicals can be smelt. We used this data to create our own dataset of 16 features and 5 classes, consisting of 5000 samples.

Since we used a synthetic dataset for this project, we could leverage the opportunity to make a highly balanced dataset. Other datasets, for e.g. SMILES, has a huge imbalance in the dataset, which affects model training.

For this project, we trained our dataset on three different models to try and gauge how different models respond to the dataset. We initially used Decision Trees and then Random Forests and then we used DNN (Deep Neural Networks) and our observations are stated further in the paper. The performance of the model for Random Forests and DNN was better than that of Decision Trees.

IV. EXPERIMENTAL DETAILS

DEEP NEURAL NETWORKS

We used a multi-layer feed forward neural network, designed for multi class classification. This model has a fully connected dense layer with 256 neurons and uses the

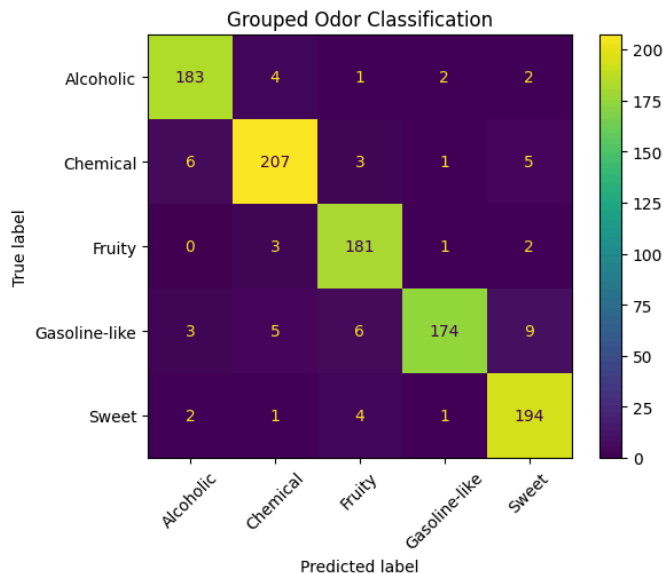
ReLU activation function.

Optimizer: Adam

Loss Function: categorical cross entropy

Layer Breakdown:

1. Dense(256, activation='relu') - First Hidden Layer with 256 neurons
2. Dropout (0.4) - Regularization to prevent overfitting
3. Dense(128, activation='relu') - Second Hidden Layer
4. Dropout(0.3) - Another Dropout Layer
5. Dense(64, activation='relu') - Third hidden layer
6. Dense (5, activation='softmax') - Output layer with 5 neurons



Classification Report:				
	precision	recall	f1-score	support
Alcoholic	0.94	0.94	0.94	192
Chemical	0.96	0.92	0.94	222
Fruity	0.93	0.97	0.95	187
Gasoline-like	0.95	0.93	0.94	197
Sweet	0.93	0.96	0.94	202
accuracy			0.94	1000
macro avg	0.94	0.94	0.94	1000
weighted avg	0.94	0.94	0.94	1000

Class: Alcoholic
True Positives: 181
False Positives: 12
False Negatives: 11
True Negatives: 796

Class: Chemical
True Positives: 204
False Positives: 9
False Negatives: 18
True Negatives: 769

Class: Fruity
True Positives: 181
False Positives: 14
False Negatives: 6
True Negatives: 799

Class: Gasoline-like
True Positives: 183
False Positives: 9
False Negatives: 14
True Negatives: 794

Class: Sweet
True Positives: 193
False Positives: 14
False Negatives: 9
True Negatives: 784

RANDOM FORESTS

The Random Forests is an ensemble machine learning algorithm that builds multiple decision trees and merges them to produce more accurate and stable predictions.

Key Parameters:

n_estimators=100 : 100 decision trees were built
criterion='gini' : Used gini impurity to determine the best splits

80% of the dataset was used to train the model and 20% was used for testing. This random forest model performed well across all odor classes, with minimal variation in precision, recall and F1-score. In terms of the prediction quality, all classes are well balanced. No class appears to be significantly underrepresented or poorly predicted, indicating that the model isn't biased or overfitted to any particular class.

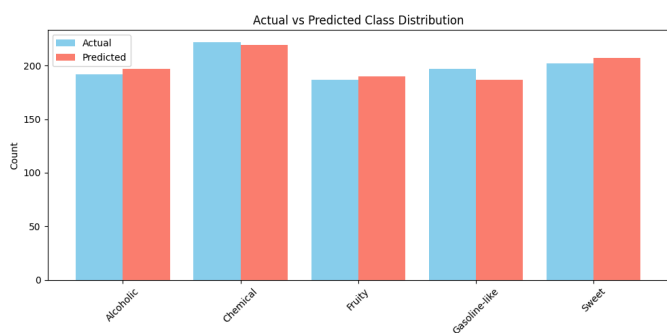
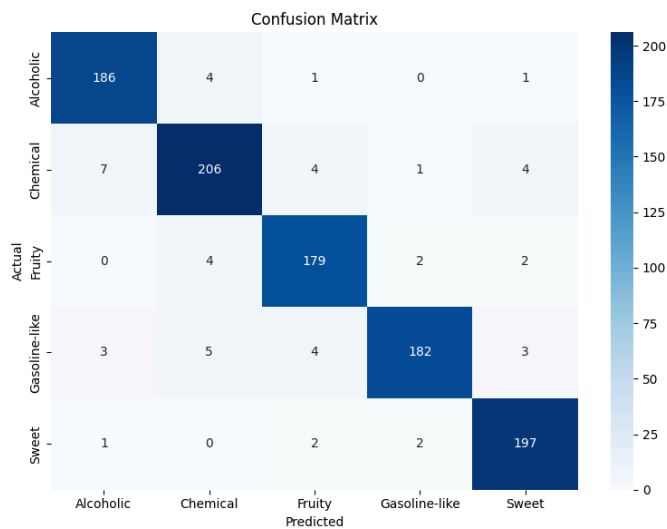
Advantages of Using Random Forest:

1. Naturally handles non-linear relationships and feature interaction.

2. Robust against overfitting due to ensemble averaging.
3. Handles both numerical and categorical data (after encoding)
4. Feature importance can be easily extracted for interpretability.

Model Accuracy: 0.95
Classification Report:

	precision	recall	f1-score	support
0	0.94	0.97	0.96	192
1	0.94	0.93	0.93	222
2	0.94	0.96	0.95	187
3	0.97	0.92	0.95	197
4	0.95	0.98	0.96	202
accuracy			0.95	1000
macro avg	0.95	0.95	0.95	1000
weighted avg	0.95	0.95	0.95	1000



DECISION TREES

A Decision Tree is a supervised machine learning algorithm used for both classification and regression tasks. For this project, this Decision Tree model uses the entropy criterion i.e. information gain and tries to reduce the uncertainty at each split. The max depth of the tree is limited to 5, which prevents the tree from growing too deep helping it prevent overfitting. For preprocessing we used Label Encoding for categorical target labels.

Pros of using Decision Trees:

1. Easy to interpret

2. No need for feature scaling
3. Captures non linear relationships
4. Handles both numerical and categorical data effectively

Cons of using Decision Trees:

1. Prone to overfitting, especially with deep trees.
2. Sensitive to small changes in data (high variance)
3. May not capture complex features or interactions well.

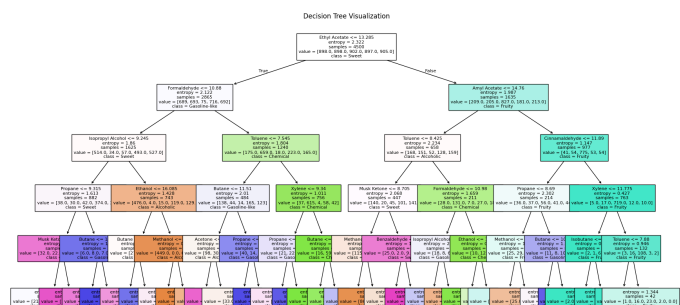
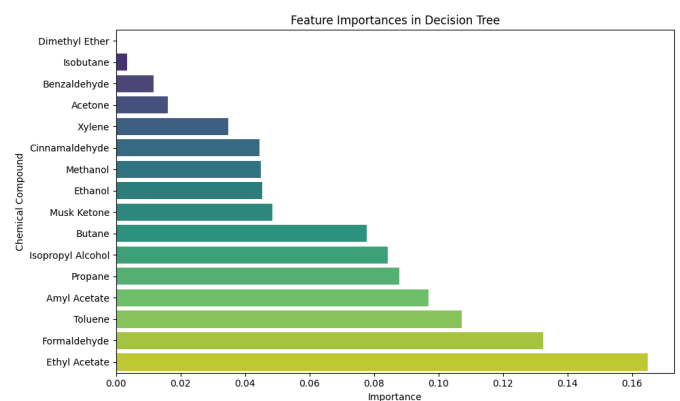
Using Random Forests is slightly better than using Decision Trees and Random Forests are ensemble models, i.e. it trains multiple decision trees on different subsets of data and aggregates their predictions (via majority voting) for classification.

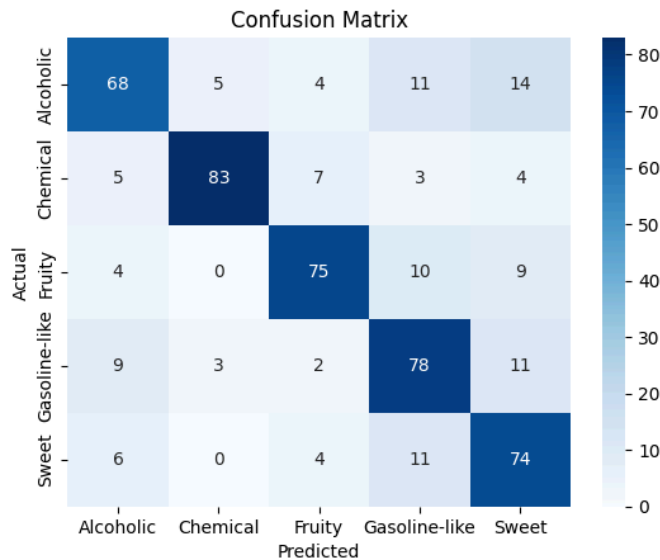
Model Accuracy: 0.76
Classification Report:

	precision	recall	f1-score	support
0	0.74	0.67	0.70	102
1	0.91	0.81	0.86	102
2	0.82	0.77	0.79	98
3	0.69	0.76	0.72	103
4	0.66	0.78	0.71	95
accuracy			0.76	500
macro avg	0.76	0.76	0.76	500
weighted avg	0.76	0.76	0.76	500

Confusion Matrix:

	Alcoholic	Chemical	Fruity	Gasoline-like	Sweet
Alcoholic	68	5	4	11	14
Chemical	5	83	7	3	4
Fruity	4	0	75	10	9
Gasoline-like	9	3	2	78	11
Sweet	6	0	4	11	74





V. SENSOR DETAILS

MQ SERIES

Table 1
Summary of compounds detected by each MQ sensor.

Type of MQ	Compound Detection	Methane	Butane	Liquid Petroleum Gas (LPG)	Smoke	Alcohol	Ethanol	Compressed Natural Gas (CNG)	Natural Gas	Carbon Monoxide (CO)	Hydrogen (H)
MQ2		X	X	X							
MQ3					X	X	X				
MQ4		X						X			
MQ5				X					X		
MQ6			X	X						X	
MQ7											X
MQ8											

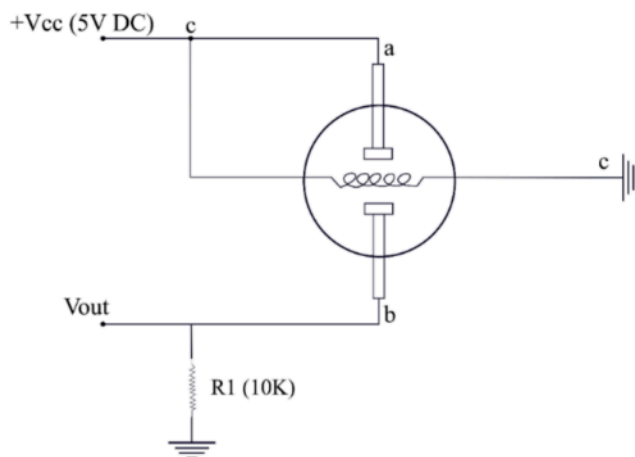


Fig. 4. Pictorial description of the MQ sensor system.

The MQ sensor series is a family of low-cost gas sensors widely used for detecting gases, and volatile compounds in the environment. Each MQ sensor can be calibrated to be more sensitive to a particular group of gases. These sensors are popular in hobby electronics, embedded systems and projects for gas leakage detection, air quality monitoring and odor analysis.

MQ sensors operate based on chemiresistive sensing. They use a metal oxide semiconductor, (typically SnO₂) whose electrical resistance changes in the presence of target gases.

Heater Coil: Maintains the sensing element at a high temperature (~300 to 500 degree celsius) which is necessary for gas sensing.

Sensing Layer: Absorbs oxygen molecules from air which capture electrons and increase resistance.

Gas Reaction: When target gases are present, they react with the oxygen on the surface, releasing trapped electrons and reducing the resistance.

Resistance Change: The sensor outputs an analog voltage value that corresponds to the resistance change, indicating gas concentration.

TGS SERIES

The Figaro TGS sensors are thick-film metal oxide semiconductor (MOS) gas sensors, typically using Tin Dioxide as the sensing material. They detect gases by measuring changes in electrical resistance.

These sensors are known for their low cost, long operational life and good sensitivity to a variety of target gases, particularly toxic and explosive ones. Sensor resistance typically shows a linear relationship with gas concentration when plotted on a logarithmic scale within a specific range.

The sensor performance is dependent on ambient temperature, humidity and partial pressure of oxygen. They require ambient oxygen (~21%) to function correctly. TGS sensors require constant regulated heater voltage for optimal performance and exhibit an initial resistance drop ("Initial Action") upon power-up that needs to be managed in the circuit design.

These sensors are also very sensitive to contaminants like silicone vapors. When turned on, they show a quick change in readings at first, and can be damaged by things like silicone vapors.

Perfume classification relies on detecting the unique chemical "fingerprint" of each scent family. TGS sensors achieve this by changing their electrical resistance when perfume VOCs interact with their surface.

No.	Sensor name	Target Gasses
1	TGS 880	Cooking Vapors
2	TGS 2620	Organic solvents
3	TGS 825	Toxic gasses
4	TGS 2602	Indoor Pollutants
5	TGS 826	Toxic gasses
6	TGS 2104	Automobile Ventilation
7	TGS 830	Chlorofluorocarbons
8	TGS 2610	Combustible gasses

CONCLUSION

This project demonstrates the feasibility and effectiveness of using synthetic datasets and machine learning models for odor classification based on sensor readings. By leveraging the sensitivity of MQ and TGS gas sensors and combining them with advanced classification algorithms, we were able to distinguish between multiple odor classes with a high degree of accuracy. Among the models, Deep Neural Networks and Random Forests showed superior performance. Furthermore, the use of a synthetic dataset allowed us to bypass issues such as class imbalance, which is commonly faced in real-world datasets such as SMILES.

Our findings lay the groundwork for future development of low-cost, scalable e-nose systems that can be deployed across a range of applications. Future work can involve real-world sensor integration, cross-domain datasets and enhanced model explainability for practical use cases.

REFERENCES

- [1] Occupational Safety and Health Administration (OSHA), "Isobutane - Chemical Data," U.S. Department of Labor, [Online]. Available: <https://www.osha.gov/chemicaldata/577>.
- [2] National Institute for Occupational Safety and Health (NIOSH), "Methanol," NIOSH Pocket Guide to Chemical Hazards, Centers for Disease Control and Prevention (CDC), [Online]. Available: <https://www.cdc.gov/niosh/npg/npgd0397.html>.
- [3] Carl Roth GmbH + Co. KG, "Cinnamaldehyde Safety Data Sheet," Carl Roth, [Online]. Available: <https://www.carlroth.com/medias/SDB-3618-GB-EN.pdf?context=bWFzdGVyfHNIY3VvaXR5RGF0YXNoZWV0c3wzMzc4Mzl8YXBwbGljYXRpb24vcGRmfGMvVmpkWEpwZEhsRVlYUmhjMmhsWlhSekwyZzROaTlvTXprdk9Ea3hNemd4TVRZek5qSTFOQzV3WkdZfGU4Nzk5ZTVjZmMxZmNkNjY3OTc3NjVIMDFjNjk0NWZjNjQxOGZkNTY3YWFEwNzIzYzAwZThlNzg4ZDAzNmExZTQ>.
- [4] **U.S. EPA**, *Odor Thresholds for Chemicals with Established Health Standards*, 2nd ed., U.S. Environmental Protection Agency, Office of Research and Development, Washington, DC, EPA/600/R-92/047, 1992.
- [5] A. Sharma, R. Kumar, S. Ranjta, and P. K. Varadwaj, "SMILES to Smell: Decoding the Structure–Odor Relationship of Chemical Compounds Using the Deep Neural Network Approach," *J. Chem. Inf. Model.*, vol. 61, no. 2, pp. 676–688, Feb. 2021, doi: 10.1021/acs.jcim.0c01288.
- [6] K. Saini and V. Ramanathan, "Predicting odor from molecular structure: A multi-label classification approach," *Scientific Reports*, vol. 12, no. 1, p. 13863, Aug. 2022, doi: 10.1038/s41598-022-18086-y.
- [7] **B. Sanchez-Lengeling, J. N. Wei, B. K. Lee, R. C. Gerkin, A. Aspuru-Guzik, and A. B. Wiltschko**, "Machine Learning for Scent: Learning Generalizable Perceptual Representations of Small Molecules," *arXiv preprint arXiv:1910.10685*, Oct. 2019. [Online]. Available: <https://arxiv.org/abs/1910.10685>
- [8] **F. J. D. Blasco, S. Viciano-Tudela, L. Parra, A. Ahmad, V. Chaloupková, R. Bados, L. S. Esteban Pascual, I. Mediavilla, S. Sendra, and J. Lloret**, "Employment of MQ gas sensors for the classification of *Cistus ladanifer* essential oils," *Sensors and Actuators B: Chemical*, vol. 367, p. 132084, 2022.
- [9] **S. Güney, A. Atasoy, and R. Burget**, "Electronic nose odor classification with advanced decision tree structures," in *Proc. 2020 International Conference on Engineering and Technology (ICET)*, Trabzon, Turkey, 2020, pp. 1–6.